

Department of Computer Science and Engineering
Jessore University of Science & Technology
1st Year 2nd Semester Final Exam-2023 (Session: 2021-22)

Course Code: CSE 1201

Total Time: 3.00 hours

Course Title: Data Structures

Total Marks: 72.0

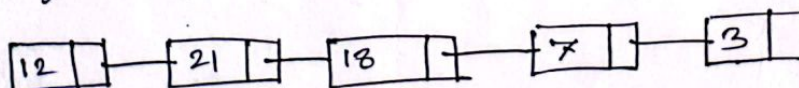
(Note: Answer any six set of questions from the following eight set of questions)

1. a) Why are data structures and algorithms so important in computer science? 2
b) Suppose a five-dimensional array A is declared using A (2:8, -4:1, 6:10, 3:7, -2:6). 4
Consider the element A [3, -1, 8, 4, 2] in A. Find the effective indices E1, E2, E3, E4, E5 and the address of the element in row-major order and column-major order, assuming Base (A) = 400 and there are w = 4 words per memory location.
c) What do you mean by complexity of an algorithm? Find the average case time complexity of linear search algorithm executed on an array contains n elements. 3
d) What do you mean by priority queue? Write down the operations of the priority queue. Also write down the program to max heapify of an array 3
2. a) How can you minimize overflow in stack? Explain with example(s). 4
b) Write a recursive procedure to solve the 'Tower of Hanoi' problem. 4
c) Consider the following queue where QUEUE is allocated 6 memory cells: 4
FRONT=2, REAR=3, QUEUE:

	B	C			
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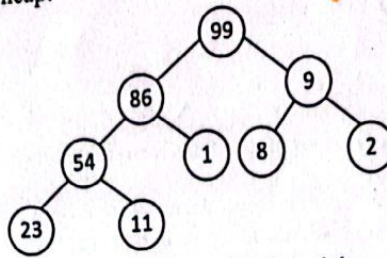
Now describe the queue including front and rear, as the following operations take place: (i) D and E inserted, (ii) B and C deleted.
3. a) Consider the following deque of characters where DEQUE is a circular array which 4
allocates six memory cells:
LEFT = 2, RIGHT = 4, DEQUE: -----, A, C, D, -----, -----.
Describe the deque while the following operations take place:
i) F is added to the right of the deque.
ii) Two letters on the right are deleted.
iii) K, L, M are added to the left of the deque.
iv) One letter on the left is deleted.
v) R is added to the left of the deque.
b) Write down the counting sort algorithm. 3
c) Consider a hash table of size 7 with hash function $h(k) = k \bmod 7$. Draw the table 5
that results after inserting in the given order, the values, 19, 26, 13, 48, 17 for each of the scenarios below:
i) When collisions are handled by separate chaining.
ii) When collisions are handled by linear probing.
iii) When collisions are handled by double hashing.
Using a second hash function $h'(k) = 5 - (k \bmod 5)$.
4. a) Compress the sentence "Happy hip hop" using Huffman coding scheme and also 5
show the effectiveness of the compression.
b) Define a Linked List. Write a function to insert an element in the middle of a Linked 4
List. Let consider the following linked list with 12 as the head. Now write an algorithm to insert a new node (15) in the middle of node 18 and 7.
c) Suppose STR1 and STR2 are character strings in memory. Write a procedure which 3
creates a new string STR3 which is the concatenation of STR1 and STR2.
5. a) Using the definition of Ackerman functions find the value of A (1, 3) and show each 4
call and return values.
b) Define maxheap. Show the maxheap building process from the following list of 4
numbers.
44, 30, 50, 22 60, 55, 77, 55
c) Prove that the height of a complete binary tree with N node is about $\log_2 N$. 4

4(b)



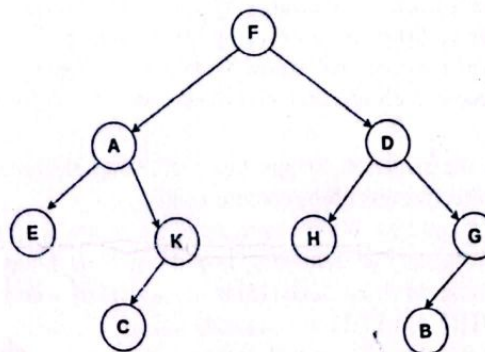
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6. a) What are the differences between heap and BST? When to use a heap and when to use a BST? 3
b) Consider the following heap: 6



For each of the following questions use the original heap. Do not use the heap obtained after each answer.

- i) Represent the heap as an array indexed by 1 through 9.
ii) Delete the element 99 from the heap shown in the above figure.
iii) Delete the element 23 from the heap shown in the above figure.
iv) Delete the element 8 from the heap shown in the above figure.
v) Is it max heap or min heap? Sorting the elements in descending order by heap sort.
- c) Discuss the advantages, if any, of a two-way list over a one-way list for each of the following operations: 3
i) Traversing the list to process each node.
ii) Deleting a node whose location LOC is given.
iii) Inserting a node after the node with a given location LOC.
7. a) Write down the output of this code with THREE pushes and ONE pop 3
`IntStack s = new IntStack( );`
`s.push(1);`
`s.push(2);`
`s.push(3);`
`s.push(12);`
`s.push(14);`
`printf(s.pop( ));`
- b) Write down the procedure of ENQUEUE and DEQUEUE using an array. 3
c) Draw a picture of the sequence 13, 4, 8, 19, 5, 11 stored as a doubly linked list using the multiple array representation. Do the same for the single array representation. 3
d) Define divide and conquer. Solve the $7^{37} \% 9$ expression using divide and conquer. 3
8. a) Consider the following graph 6



- list the nodes of the tree in preorder, postorder and inorder
b) Build a heap with following list of numbers 44, 30, 50, 25, 60, 55, 77, 55 *min heap*

Course Code: CSE 1203

Course Title: Digital Logic Design

Time: 3.00 hours
Marks: 72

[N.B. Answer any six sets of questions from the following eight sets of questions.]

- 1 a) Represent the decimal number 8620 in
i. BCD
ii. Excess-3
iii. Gray code
iv. Binary number 6
- b) What is the largest binary number that can be expressed with 12 bits? What are the equivalent decimal and hexadecimal numbers? 3
- c) Perform subtraction on (2043-6152) using the 10's complement of the subtrahend where the result should be negative. Find its 10's complement and affix a minus sign. Verify your answer. 3
- 2 a) Reduce the following Boolean expression to the required number of literals- 6
i. $ABC + A'B'C + A'BC + ABC' + A'B'C'$ to five literals
ii. $[(CD)' + A]' + A + CD + AB$ to three literals
iii. $(A + C + D)(A + C + D')(A + C + D)(A + B')$ to four literals
- b) Given the Boolean function: $F = xy + x'y' + y'z$ 4
i. Implement it with only OR and NOT gates
ii. Implement it with only AND and NOT gates
- c) Convert the following to the other canonical form 2
i. $F(A, B, C, D) = \sum(0, 2, 6, 11, 13, 14)$
ii. $F(A, B, C, D) = \prod(0, 1, 2, 3, 4, 6, 12)$
- 3 a) Obtain the simplified expression in the sum of products 4
i. $(A' + B' + D)(A' + D')(A + B + D')(A + B' + C + D)$
ii. $x'z' + y'z' + yz' + xyz$
- b) Write the Boolean equations and draw the logic diagram of the circuit whose outputs are defined by the following truth table. 6

f_1	f_2	a	b	c
1	0	0	0	0
0	0	0	0	1
0	1	0	1	0
1	1	0	1	1
0	1	1	0	0
0	1	1	0	1
1	1	1	1	0
1	0	1	1	1

- c) Simplify $f(w, x, y, z) = \sum(0, 1, 2, 8, 12, 13, 14) + d(3, 5, 10, 15)$ 2

- 4 a) Design a combinational circuit that takes a two-bit number and generates the cube of the input number as output. 4
- b) Implement a combinational circuit that converts a 4-bit excess-3 code to gray code 6
- c) What are universal gates? Why are they called so? 2
- 5 a) Prove that $x \oplus 1 = x'$ and $x \oplus 0 = x$. 4
- b) Design the BCD-to-decimal decoder to give an output of all 0s when any invalid input combination occurs. 8
- 6 a) Differentiate between multiplexer and Decoder 2
- b) Design a BCD counter with JK flip-flops. 6
- c) Write short notes on 4
- Ring counter
 - Parallel in Serial Out (PISO) Shift register
- 7 a) Describe the truth table of an SR flip-flop implemented with NOR. 3
- b) Define Race Around condition in J-K flip-flop. How can one resolve this issue? Explain with appropriate figure and waveforms. 6
- c) Show the logic diagram of a clocked D flip-flop with AND and NOR gates. 3
- 8 a) Design a full Adder with 4
- Half adders
 - The sum of products form
- b) Demonstrate the state diagram of a 4-bit binary counter. 4
- c) Illustrate the block diagram of a memory unit showing communication with the environment. 4



Department of Computer Science and Engineering
Jashore University of Science and Technology

Semester Final Examination-2023
B.Sc. (Engg.) 1st Year 2nd Semester, Session: 2021-22

Course Code: EEE-1201

Course Title: Electrical Devices and Circuits

Time: 3.00 hours

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. (a) At a temperature of 27°C (common temperature for components in an enclosed operating system), determine the thermal voltage V_T 4
(b) Prove that - "The higher the current through a diode, the lower is the dc resistance level." 4
(c) Compare conductor, insulators and semiconductor using band theory 4
2. (a) Determine V_o and I_D for the networks of Fig 1 3+3

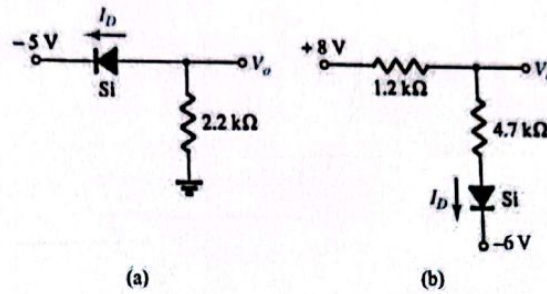


Fig. 1

- (b) Determine V_o and I_D for the networks of Fig 2 3+3

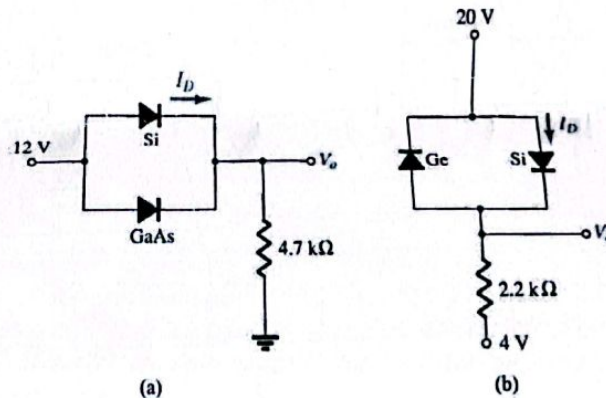


Fig. 2

3. (a) Determine the output waveform for the network of Fig. 3 and calculate the output dc level and the required PIV of each diode. 6

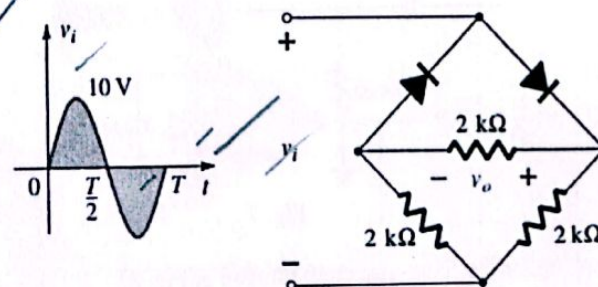


Fig. 3

- (b) For the Zener diode network of Fig. 4, determine V_L , V_R , I_Z , and P_Z

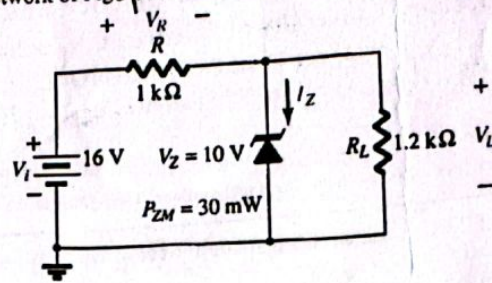


Fig. 4

4. (a) Find the output voltage for the following network if the applied signal is the square wave as indicated in Fig 5

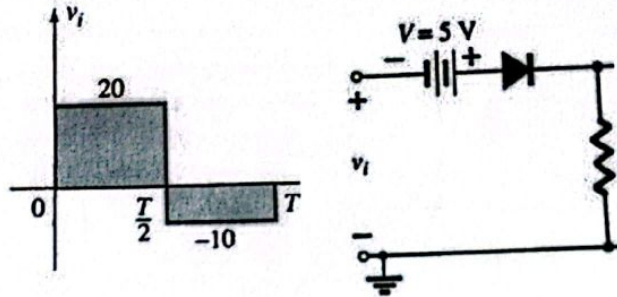


Fig. 5

- (b) Determine v_o for the network of Fig. 6 for the input indicated

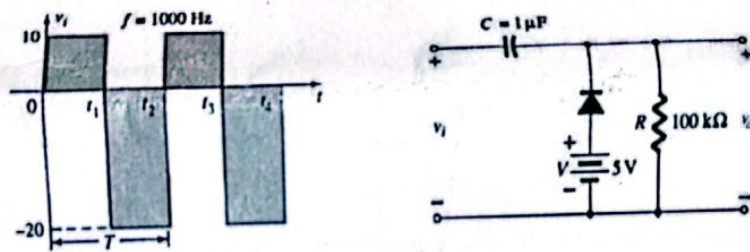


Fig. 6

5. (a) Determine the dc bias voltage V_{CE} and the current I_C for the voltage divider configuration of Fig. 7.

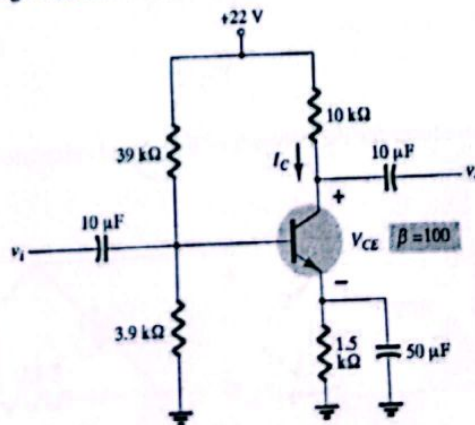


Fig. 7

- (b) Determine the following for the fixed-bias configuration of Fig. 8.

- I. I_{BQ} and I_{CQ}
- II. V_{CEQ}
- III. V_B and V_C
- IV. V_{BC}

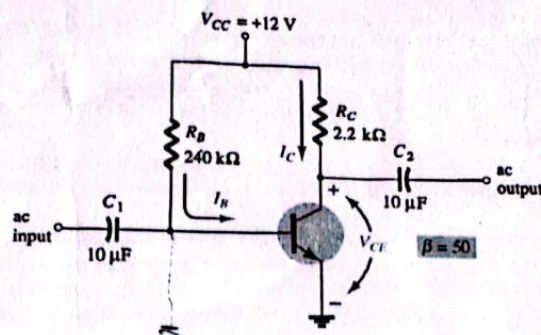


Fig: 8

6. (a) Draw the ac equivalent circuit, the hybrid equivalent model and the approximate hybrid equivalent model of the following BJT circuit in Fig: 9 4+4+2

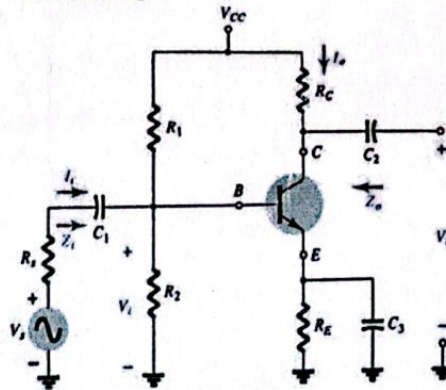


Fig: 9

- (b) Explain Early effect in BJT 2
7. (a) Semiconductor devices collectively contribute to the advancement of intelligent systems and technologies in the 5th Industrial Revolution. While BJT devices excel in high-current applications, JFETs cater to low-noise and high-impedance requirements, and MOSFETs thrive in power electronics and efficient energy conversion. Their unique capabilities enable seamless connectivity, data processing, power management, and signal amplification, making them indispensable components in the implementation of AI, IoT, robotics, and various other transformative technologies that drive the 5IR forward. Specific application of biasing method, choosing right configuration are essential for real life circuit design for their stability. Now, Describe the steps of designing a real-life transistor biasing circuit for low powered transistor 8
- (b) "E-MOSFET operates only in the enhancement mode"- Justify the statement by describing the working principle of an E-MOSFET. 4
8. (a) Explain voltage transfer curve of op-amp 4
- (b) Calculate the output voltage of a noninverting amplifier (as in Fig. 10) for values of $V_1 = 2V$, $R_f = 500k\Omega$, and $R_1 = 100 k\Omega$ 4

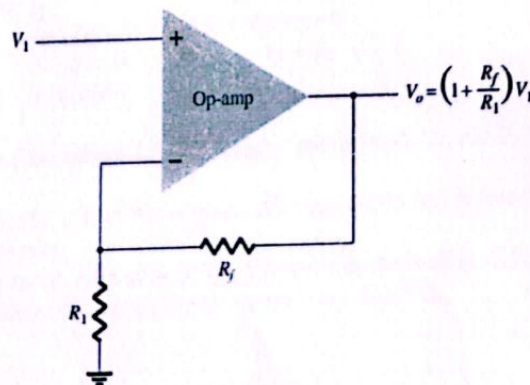


Fig: 10

- (c) Explain the working principle of op-amp differentiator 4



Course Code: MATH 1201

Course Title: Differential Equations and Geometry

Time: 3.00 hours

Marks: 72

[N.B. Answer any 6 questions from following 8 questions. The figures in the margin indicate full marks.]

1. a) Find a solution to the boundary-value problem $y'' + 4y = 0$; $y(\pi/8) = 0$, $y(6) = 1$, if the general solution to the differential equation is $y(x) = c_1 \sin 2x + c_2 \cos 2x$. 04
 b) Write the differential equation $y(yy' - 1) = x$ in two different differential form. 04
 c) Determine whether the differential equation $y' = y/x$ is exact. 04

2. a) Solve the equation $\frac{dy}{dx} = \frac{x^2+2}{y}$ 04

b) Solve $y' = \frac{2y^4+x^4}{xy^3}$ 04

c) Solve $(y - xy^2)dx + (x + x^2y^2)dy = 0$ 04

3. a) Find an integrating factor for the following equations: 04

i) $y' - 2xy = x$

ii) $y' + (4/x)y = x^4$

b) Solve $\frac{dQ}{dt} + \frac{2}{10+2t}Q = 4$ 04

c) Solve the equation: 04

$$(1+x^2)\frac{dy}{dx} + 2xy = 4x^2$$

4. a) Solve $\ddot{y} - 6\dot{y} + 25y = 2\sin\frac{t}{2} - \cos\frac{t}{2}$ 04

b) Solve $\ddot{x} + 4x = \sin^2 2t$; $x(0) = 0$, $\dot{x}(0) = 0$ 04

c) Solve $\frac{dT}{dt} + kT = 100k$, where k denotes a constant. 04

5. a) Show that the four points $(0, 4, 3)$, $(-1, -5, -3)$, $(-2, -2, 1)$ and $(1, 1, -1)$ are coplanar and find the common plane. 04

- b) Find the equation of the plane through the intersection of planes $2x - y = 0$ and $3z - y = 0$ and perpendicular to plane $4x + 5y - 3z = 8$. 04

- c) Find the equation of the plane through the intersection of the planes $x - 2y + 3z + 4 = 0$ and $2x - 3y + 4z - 7 = 0$ and the point $(1, -1, 1)$ 04

6. a) Find the angle between the straight lines represented by the equation $ax^2 + 2hxy + by^2 = 0$. Also find the condition that the two lines will be coincident and are at right angles. 06

- b) Prove that the lines joining the origin to the intersections of $ax^2 + 2hxy + by^2 + 2gx = 0$ and $a'x^2 + 2h'xy + b'y^2 + 2g'x = 0$ are at right angles if $\frac{a+b}{g} = \frac{a'+b'}{g'}$ 06

7. a) Find the equation of the line through the point $(1, 2, 3)$ and parallel to the line $x - y + 2z = 5$, $3x + y + z = 6$. 04

- b) Find the equation of the plane through the line $3x - 4y + 5z = 10$, $2x + 2y - 3z = 4$ and parallel to $x = 2y = 3z$. 04

- c) Prove that the lines $2x + 3y - 4z = 0$, $3x - 4y + z = 7$ and $5x - y - 3z + 12 = 0$, $x - 7y + 5z - 6 = 0$ are parallel. 04

8. a) Find the Centre of the sphere through the four points $(4, -1, 2)$, $(0, -2, 3)$, $(1, -5, 1)$ and $(2, 0, 1)$ 04

- b) Find the equation of a sphere which passes through origin and makes equal intercepts of unit length on the axes. 04

- c) A plane passes through a fixed point (p, q, r) and cuts the axes in A, B, C. Show that the locus of the Centre of the sphere OABC is $p/x + q/y + r/z = 2$. 04

Department of Computer Science and Engineering
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1st Year 2nd Semester Examination, Session 2021-22

Course Title: Physics for Computer Engineers
Course Code: PHY 1201

Time: 3 hours
Full Marks: 72

N.B. There are eight questions, answer any six questions. The figures shown in the margin indicate full marks

- 1 (a) On the basis of kinetic theory of gases, give the kinetic interpretation of temperature. 3
- (b) State the Maxwell's law of equipartition of energy. At high temperature, find out the degrees of freedom for triatomic molecule (H_2S), when it is in a non-linear structure. 4
- (c) What is the aim of Andrew's experiment? For CO_2 gas, draw isothermals and discuss the phenomenon of below and above critical isothermals. 3
- (d) Van der Waal's constants for a gas are $a = 6.9 \times 10^{-2} Jm^3mole^{-2}$ and $b = 2.9 \times 10^{-5} m^3mole^{-1}$. The universal gas constant $R = 8.31 Jmole^{-1}K^{-1}$. Calculate the critical temperature of the gas. 2
- 2 (a) Define internal energy and heat 2
- (b) What is the energy required to raise the temperature of 0.500 kg of water by $3.00^\circ C$? Specific heat of water is $4186 J/(kg^\circ C)$ 4
- (c) A 0.0500-kg ingot of metal is heated to $200.0^\circ C$ and then dropped into a calorimeter containing 0.400 kg of water initially at $20.0^\circ C$. The final equilibrium temperature of the mixed system is $22.4^\circ C$. Find the specific heat of the metal. 4
- (d) Define latent heat of a substance 2
- 3 (a) Outline the fundamental assumptions of the kinetic theory of gases, which serves as the basis for understanding the behavior of gases at the molecular level. 5
- (b) What is the critical point and critical constant of a substance? Define different critical constants. 5
- (c) What do you mean by Reversible and Irreversible Processes? 2
- 4 (a) An engine transfers $2.00 \times 10^{-3} J$ of energy from a hot reservoir during a cycle and transfers $1.50 \times 10^{-3} J$ as exhaust to a cold reservoir. Find the efficiency of the engine. How much work does this engine do in one cycle? 4
- (b) Describe Carnot engine. 5
- (c) State Kelvin-Planck form of the second law of thermodynamics 3

- 5 (a) Define wavelength, amplitude of a wave with image 2
- (b) What is wave reflection? Describe free end reflection and fixed end reflection 6
- (c) Describe constructive and destructive interference with image 4
- 6 (a) State the superposition principle of wave and resonance frequency 4
- (b) State and explain doppler effect 6
- ~~(c)~~ What is the decibel level difference between two sounds if one sound has an intensity that is 100 times greater than the other sound? 2
- 7 (a) A 200-g block connected to a light spring for which the force constant is 5.00 N/m is free to oscillate on a horizontal, frictionless surface. The block is displaced 5.00 cm from equilibrium and released from rest. Find period of the motion, maximum speed and maximum acceleration of the block 6
- ~~(b)~~ Explain the factors that affect the period of a pendulum's oscillation. Provide a mathematical expression for the period of a simple pendulum in terms of its length 'L' and the acceleration due to gravity 'g.' Include any relevant assumptions or conditions in your explanation. 6
- ~~8 (a)~~ Define the relationship between average coefficient of linear expansion and average coefficient of volumetric expansion 6
- ~~(b)~~ A 1.00-liter container holds a sample of gas at a pressure of 2.00 atm and a temperature of 300 K. If the pressure is increased to 3.00 atm while keeping the volume constant, what will be the new temperature of the gas in degrees Celsius? 4
- (c) Define heat capacity of a substance 2